

Feature Extraction Techniques for E-commerce Image Classification

CSE 802 Project Proposal: Roshni Ranjita Bhowmik

1. Problem Domain

With the rapid growth of online retail platforms, automated product categorization has become a critical component of e-commerce systems. Accurate classification of product images into predefined categories improves search functionality, recommendation systems, and inventory organization. However, image classification performance heavily depends on how visual information is represented.

This project addresses a pattern classification problem in computer vision, focusing on identifying the most effective feature extraction techniques for classifying product images. Specifically, the project evaluates how different feature representations such as raw pixel values, shape-based descriptors, texture-based descriptors, and color-based features impact classification accuracy in a multi-class setting.

The goal is not only to build a classifier but also to gain insights into which types of visual features are most informative for real-world image classification tasks.

2. Dataset(s) that will be Used

Dataset Name: Fashion Product Images (Small) Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/paramaggarwal/fashion-product-images-small/data?se>

Dataset Description

This dataset contains real-world fashion product images collected from an e-commerce platform. It includes metadata such as product category, subcategory, gender, and color.

Purpose of the Dataset

The dataset will be used to train and evaluate machine learning models for multi-class image classification. It is well-suited for this project due to its real-world relevance and availability of labeled data.

Input Features

The input consists of image data, which will be transformed into different feature representations: Raw pixel values (flattened vectors), HOG (Histogram of Oriented Gradients), ORB (Oriented FAST and Rotated BRIEF) descriptors, Color histograms (RGB distribution)

Each representation will result in a high-dimensional feature space (greater than 50 dimensions).

Output Classes

A subset of categories will be selected to ensure a balanced multi-class problem, such as:

- Apparel
- Footwear
- Accessories
- Personal Care

3. Problem Statement and Experiments

Problem Statement

The primary objective of this project is to evaluate and compare the effectiveness of different image feature extraction techniques for multi-class classification of e-commerce product images.

Experiments

- Data Preprocessing
- Feature Extraction
- Dimensionality Reduction
- Classification Models
- Evaluation Metrics
- Experimental Analysis

Through this project, I aim to gain a deeper understanding of how different feature representations affect classification performance in real-world image data. Specifically, I am interested in exploring whether handcrafted features such as HOG, ORB, and color histograms outperform raw pixel representations, and how dimensionality reduction techniques influence both accuracy and computational efficiency.

Additionally, this project will provide insights into designing efficient image classification systems for practical applications such as e-commerce product categorization.

5. References

1. <https://www.kaggle.com/datasets/paramaggarwal/fashion-product-images-small/data?select=image>